

☒ EASY

☐ ALGEBRA

Mixed

10 questions · ~15 min

03

Q1

ALGEBRA

The function f is defined by $fx = 80 - 6x$. What is the value of $f7$?

A. 13

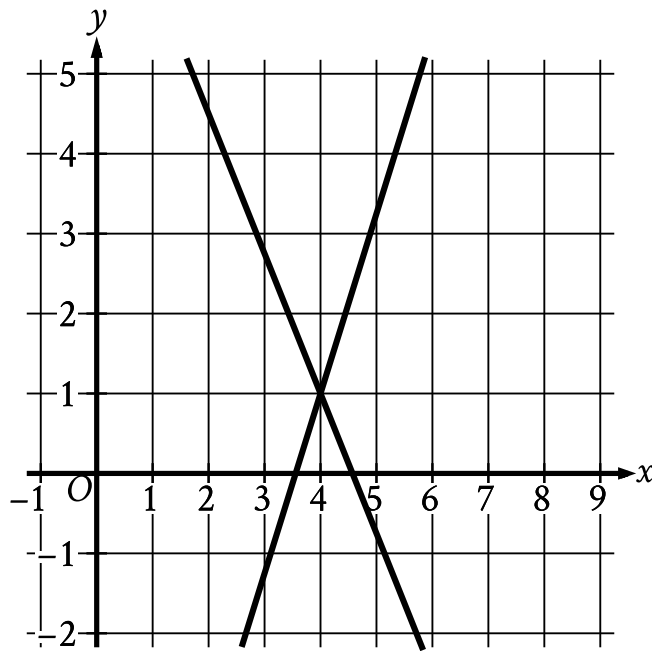
B. 38

C. 74

D. 81

The length, y , of a white whale was 162 centimeters cm when it was born and increased an average of 4.8 cm per month for the first 12 months after it was born. Which equation best represents this situation, where x is the number of months after the whale was born and y is the length, in cm, of the whale?

- A. $y = 162x$
- B. $y = 162x + 162$
- C. $y = 4.8x + 4.8$
- D. $y = 4.8x + 162$



- For the first line in the system:
 - The line slants sharply down from left to right.
 - The line passes through the following points:
 - (4 comma 1)
 - (StartFraction 32 Over 7 EndFraction comma 0)
- For the second line in the system:
 - The line slants sharply up from left to right.
 - The line passes through the following points:
 - (StartFraction 32 Over 9 EndFraction comma 0)
 - (4 comma 1)

The graph of a system of linear equations is shown. The solution to the system is x, y . What is the value of x ?

STUDENT-PRODUCED RESPONSE

	$\frac{\quad}{\quad}$	$\frac{\quad}{\quad}$	$\frac{\quad}{\quad}$
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

The function f is defined by $f(x) = 25x + 30$. What is the value of fx when $x = 2$?

- A. 50
- B. 57
- C. 80
- D. 110

Q5

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

A special camera is used for underwater ocean research. When the camera is at a depth of 58 fathoms, what is the camera's depth in feet? 1 fathom = 6 feet

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Marta has 7,500 pesos she will convert to US dollars using a currency exchange service. At this time, the currency exchange rate is 1 peso = 0.075 US dollars. The exchange service will charge Marta a 2% fee on the converted US dollar amount. How many US dollars will Marta receive from the currency exchange after the 2% fee is applied?

- A. \$551.25
- B. \$562.50
- C. \$5,625.00
- D. \$98,000.00

The high temperature, in degrees Fahrenheit ($^{\circ}\text{F}$), in a certain city was recorded for each of 5 days. The data are shown below.

Day	1	2	3	4	5
High temperature ($^{\circ}\text{F}$)	81	80	81	81	82

Over this 5-day period, which of the following is NOT equal to 81°F ?

- A. Median of the high temperatures
- B. Mean of the high temperatures
- C. Mode of the high temperatures
- D. Range of the high temperatures

Which expression is equivalent to $20w - 4w + 3w$?

A. $10w$

B. $13w$

C. $19w$

D. $21w$

$$\frac{x^2}{25} = 36$$

What is a solution to the given equation?

- A. 6
- B. 30
- C. 450
- D. 900

Q10

GEOMETRY AND TRIGONOMETRY

The area of a rectangle is 630 square inches. The length of the rectangle is 70 inches. What is the width, in inches, of this rectangle?

- A. 9
- B. 70
- C. 315
- D. 560